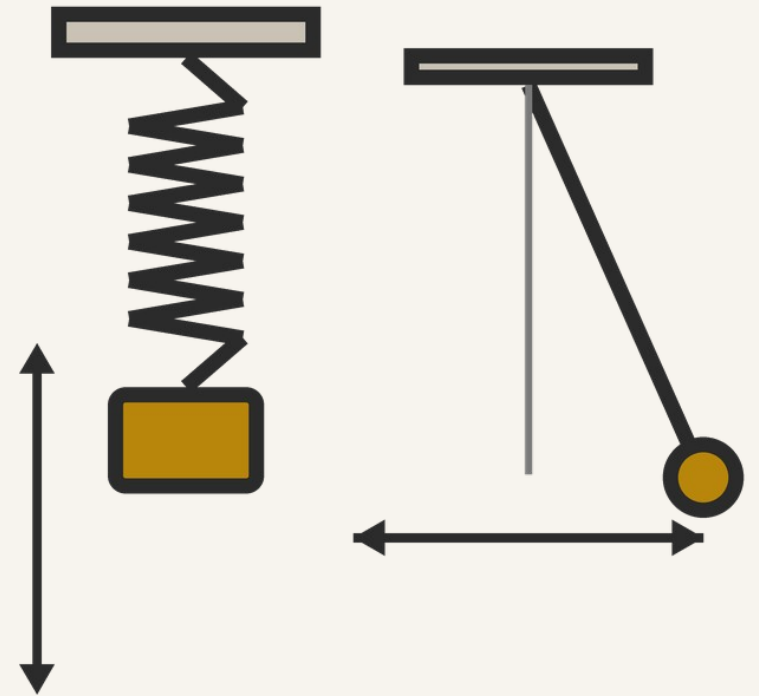


Simple Harmonic Motion

Pull it, let it go — and the maths tells you where it is at every instant.

Advanced Higher Physics



LEARNING INTENTION

To define simple harmonic motion, derive its equations of motion and energy, and use them to solve problems.

By the end of this topic...

- 1 I can state the definition of SHM from memory.
- 2 I can derive $a = -\omega^2 y$ from $y = A \sin \omega t$ and from $y = A \cos \omega t$.
- 3 I can derive $v = \pm \omega \sqrt{A^2 - y^2}$.
- 4 I can derive $E_k = \frac{1}{2} m \omega^2 (A^2 - y^2)$.
- 5 I can describe underdamping, critical damping and overdamping.

PERIOD 1 • DEFINING SHM

When is motion simple harmonic?

One condition decides it.

Systems that oscillate with SHM

Each stores potential energy and has mass to move

System	Stores E_p as	E_k of
Mass on a spring	elastic energy of the spring	the mass
Simple pendulum	gravitational PE of the bob	the bob
Trolley and two springs	elastic energy of the springs	the trolley
Tube floating in water	gravitational PE of the tube	the tube

Definition of simple harmonic motion



LEARN THIS WORD FOR WORD — asked in 2016, 2020 and 2022

The unbalanced force, and therefore the acceleration, are proportional to the displacement from the rest position and act in the opposite direction to it. $F = -ky$

Word for word from the marking instructions for 2020 Q9(a).

2020 marking instructions: do NOT accept ‘force is proportional to displacement’ without reference to direction.



The restoring force

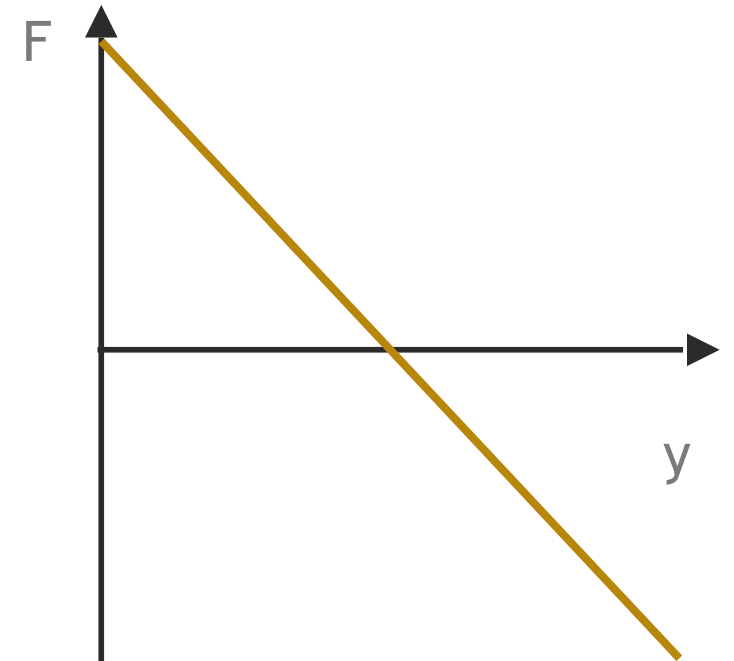
For a spring, the restoring force obeys Hooke's law:

$$F = -ky$$

k is the force constant, in N m^{-1} .

The minus sign means the force always points back

towards the rest position.



From Hooke's law to the SHM condition

JOTTER NOTES



MANDATORY DERIVATION — you must reproduce every line

$$F = -ky$$

The restoring force on the mass.

$$F = ma$$

Newton's second law for that force.

$$ma = -ky$$

One force, so equate the two.

$$a = -k/m y$$

Divide both sides by the mass m .

$$a = -\omega^2 y$$

k/m is always positive — call it ω^2 .

ω^2 is only a name for k/m . Next period shows why that name is ω .

Angular frequency



$$\omega = 2\pi f \qquad \omega = 2\pi/T$$

ω

angular
frequency, rad s^{-1}

f

frequency, Hz

T

period, s

ω is called the angular frequency in SHM, and $\omega^2 = k/m$.

Worked example 1 – 2018 Q9(a)

A ball-bearing on a track makes 1.5 oscillations in 2.5 s.
Its amplitude is 0.20 m. Find ω and the maximum speed.

$$\text{oscillations} = 1.5$$

$$t = 2.5 \text{ s}$$

$$A = 0.20 \text{ m}$$

$$\omega = ?$$

$$v_{\text{max}} = ?$$

$$T = 2.5/1.5 = 1.67 \text{ s}$$

$$\omega = 2\pi/T = 2\pi/1.67$$

$$\omega = 3.8 \text{ rad s}^{-1}$$

$$v = \pm \omega \sqrt{A^2 - y^2}$$

$$\underline{v = \pm 3.8 \times 0.20 = \pm 0.76 \text{ m s}^{-1}}$$

Starter — from memory

Close your notes. Write full sentences, not bullet points.

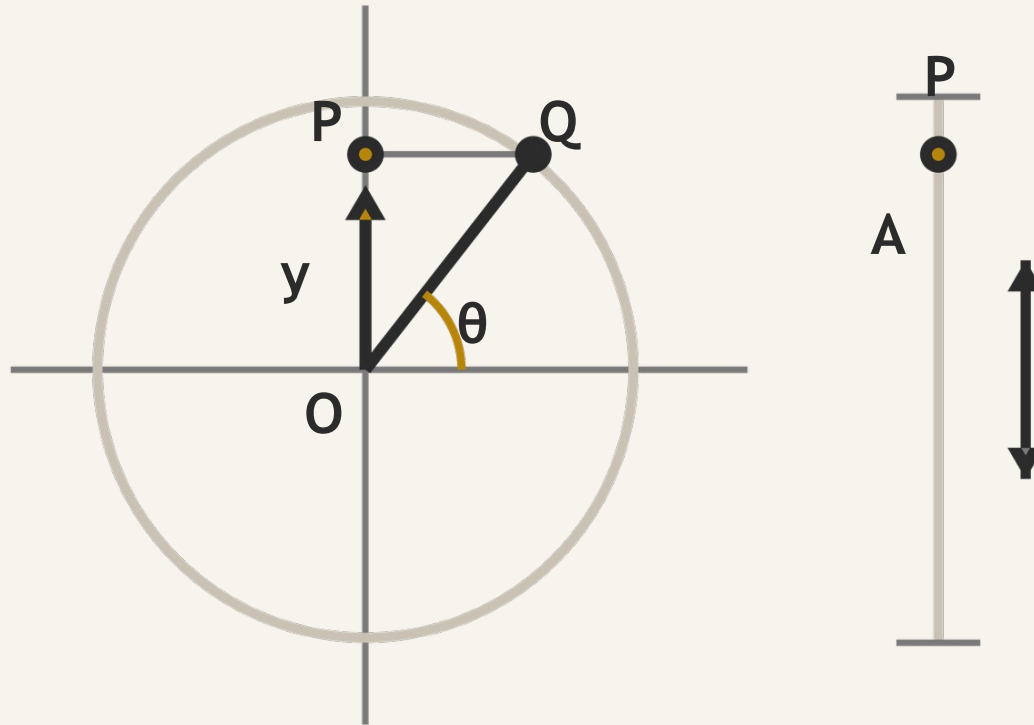
- 1 State the definition of simple harmonic motion.
- 2 Define the moment of inertia of an object.
- 3 State the principle of conservation of angular momentum.
- 4 Write ω in terms of f , and in terms of T .
- 5 Derive $v = u + at$ using calculus.

PERIOD 2 • EQUATIONS OF MOTION

Where the equations come from

A circle, seen edge-on, is simple harmonic motion.

SHM and circular motion



Q circles at constant ω ; its shadow P performs SHM, and $\theta = \omega t$.

$y = A \sin \omega t$ gives $a = -\omega^2 y$ — part 1

JOTTER NOTES



MANDATORY DERIVATION — you must reproduce every line

$$y = A \sin \omega t$$

Start from the displacement,

$$v = dy/dt$$

Velocity is the rate of change of y .

$$v = \omega A \cos \omega t$$

$\sin \omega t$ differentiates to $\omega \cos \omega t$.

Now differentiate a second time to reach the acceleration.

$y = A \sin \omega t$ gives $a = -\omega^2 y$ — part 2



MANDATORY DERIVATION — you must reproduce every line

$$a = dv/dt = (d^2 y)/(d t^2)$$

Acceleration: rate of change of v.

$$a = -\omega^2 A \sin \omega t$$

$\cos \omega t$ differentiates to $-\omega \sin \omega t$.

$$a = -\omega^2 y$$

$A \sin \omega t$ is y — substitute.

a is proportional to $-y$, so this is a solution for SHM.

$y = A \cos \omega t$ gives $a = -\omega^2 y$ — part 1

JOTTER NOTES



MANDATORY DERIVATION — you must reproduce every line

$$y = A \cos \omega t$$

Start again, with $y = A$ at $t = 0$.

$$v = dy/dt$$

The same first step: differentiate.

$$v = -\omega A \sin \omega t$$

$\cos \omega t$ differentiates to $-\omega \sin \omega t$.

Again, differentiate a second time to reach the acceleration.

$y = A \cos \omega t$ gives $a = -\omega^2 y$ — part 2



MANDATORY DERIVATION — you must reproduce every line

$$a = dv/dt = (d^2 y)/(d t^2)$$

Acceleration: rate of change of v.

$$a = -\omega^2 A \cos \omega t$$

$-\sin \omega t$ differentiates to $-\omega \cos \omega t$.

$$a = -\omega^2 y$$

$A \cos \omega t$ is y — substitute.

Use sin when $y = 0$ at $t = 0$; use cos when $y = A$ at $t = 0$.

Deriving $v = \pm\omega\sqrt{A^2 - y^2}$ — part 1

JOTTER NOTES



MANDATORY DERIVATION — you must reproduce every line

$$y = A \sin \omega t$$

The displacement, from the derivation above.

$$v = \omega A \cos \omega t$$

Its velocity, from the same derivation.

$$\sin \omega t = y/A$$

Rearrange the first equation for $\sin \omega t$.

$$\cos \omega t = v/\omega A$$

Rearrange the second equation for $\cos \omega t$.

Both are now in terms of y and v . The identity comes next.

Deriving $v = \pm\omega\sqrt{A^2 - y^2}$ — part 2



MANDATORY DERIVATION — you must reproduce every line

$$\sin^2 \omega t + \cos^2 \omega t = 1$$

$$y^2/A^2 + v^2/(\omega^2 A^2) = 1$$

$$\omega^2 y^2 + v^2 = \omega^2 A^2$$

$$v^2 = \omega^2 (A^2 - y^2)$$

$$v = \pm\omega\sqrt{A^2 - y^2}$$

The identity that links them.

Substitute both into the identity.

Multiply every term by $\omega^2 A^2$.

Subtract $\omega^2 y^2$; factorise out ω^2 .

Square root both sides. The $\pm$ is there because the mass passes each point in both directions.



Velocity at the ends and at the middle

Put $y = \pm A$ into the relationship: the square root is zero, so the mass is momentarily at rest at each extreme.

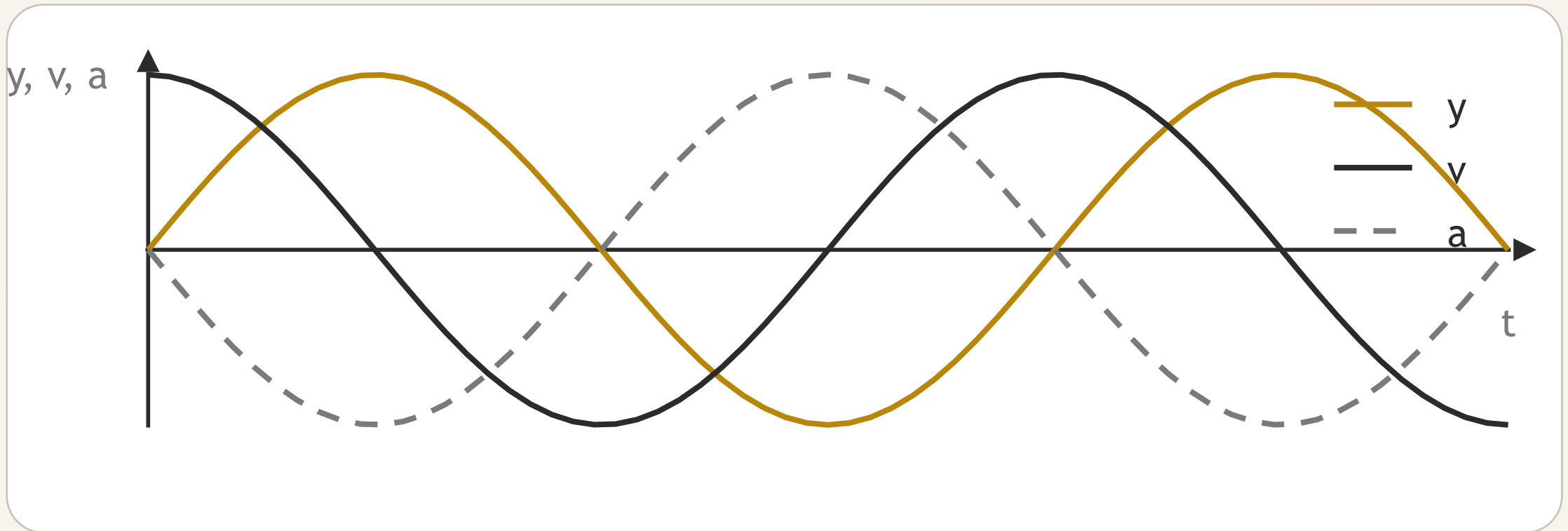
Put $y = 0$ into it: the speed is at its maximum, $v = \pm \omega A$, as the mass passes through the rest position.

Displacement, velocity and acceleration

JOTTER NOTES



v leads y by a quarter period; a is out of phase with y



2022 Q10(b) – the spring solution, part 1

JOTTER NOTES



MANDATORY DERIVATION – you must reproduce every line

$$y = A \cos (\sqrt{k/m} t)$$

The given displacement.

$$dy/dt = -A \sqrt{k/m} \sin (\sqrt{k/m} t)$$

Differentiate once.

$$(d^2 y)/(d t^2) = -k/m A \cos (\sqrt{k/m} t)$$

Differentiate again.

Both differentiations must be shown, or the marker caps it at 1.

2022 Q10(b) – the spring solution, part 2

JOTTER NOTES



MANDATORY DERIVATION – you must reproduce every line

$$(d^2 y)/(d t^2) = -k/m y$$

A cos(...) is y – substitute it into the last line of part 1.

$$m (d^2 y)/(d t^2) = -ky$$

Multiply both sides by m.

$$m (d^2 y)/(d t^2) + ky = 0$$

Add ky to both sides.

The final line must appear, or the marker caps it at 1 of 2.

Starter — from memory

Close your notes. Every step of the derivations, in order.

- 1 State the definition of simple harmonic motion.
- 2 Derive $a = -\omega^2 y$ from $y = A \sin \omega t$.
- 3 Derive $v = \pm \omega \sqrt{A^2 - y^2}$.
- 4 State where v is greatest, and where a is greatest.
- 5 Define the moment of inertia of an object.

PERIOD 3 • ENERGY AND DAMPING

Where the energy goes

Total energy is constant — until friction gets a vote.

Deriving $E_k = \frac{1}{2}m\omega^2(A^2 - y^2)$ – part 1

JOTTER NOTES



MANDATORY DERIVATION – you must reproduce every line

$$E_k = \frac{1}{2} m v^2$$

Kinetic energy of the mass.

$$v = \pm \omega \sqrt{A^2 - y^2}$$

Velocity at any displacement y .

$$v^2 = \omega^2 (A^2 - y^2)$$

Square it; the $\pm$ disappears.

Now substitute this expression for v^2 into E_k .

Deriving $E_k = \frac{1}{2}m\omega^2(A^2 - y^2)$ – part 2

JOTTER NOTES



MANDATORY DERIVATION – you must reproduce every line

$$E_k = \frac{1}{2} m v^2$$

The kinetic energy again.

$$E_k = \frac{1}{2} m \times \omega^2 (A^2 - y^2)$$

Substitute for v^2 .

$$E_k = \frac{1}{2} m \omega^2 (A^2 - y^2)$$

Tidy up – the required relationship.

E_k is greatest at $y = 0$, and zero at $y = \pm A$.



Potential energy and total energy

Take the potential energy to be zero at the rest position.

The energy the mass loses as kinetic it gains as potential:

$$E_p = \frac{1}{2} m \omega^2 y^2$$

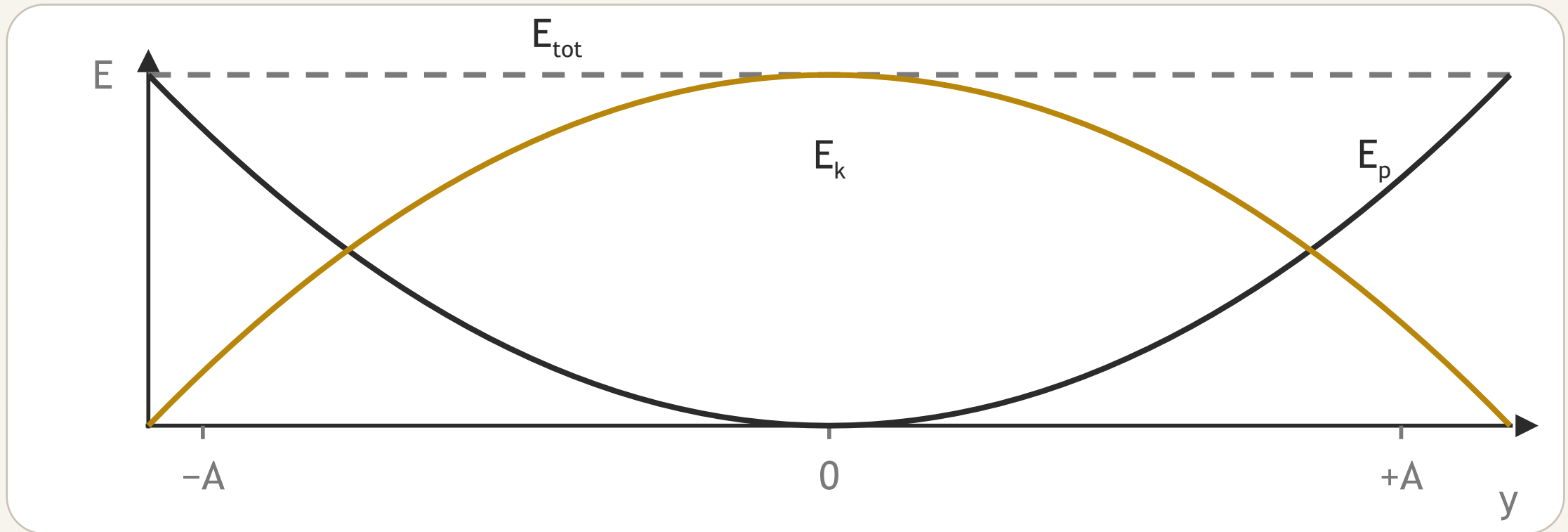
Adding E_k and E_p gives a total that does not depend on y :

$$E_{tot} = \frac{1}{2} m \omega^2 A^2$$

Energy against displacement



The two parabolas always add to the same total



Worked example 3 – 2019 Q11(a)

A tin of paint, mass 3.67 kg, is shaken 580 times per minute. Its amplitude is 0.013 m. Find ω and the maximum E_k .

$$m = 3.67 \text{ kg}$$

$$f = 580 \text{ per minute}$$

$$A = 0.013 \text{ m}$$

$$\omega = ?$$

$$E_k = ?$$

$$f = 580/60 = 9.67 \text{ Hz}$$

$$\omega = 2\pi f = 2\pi \times 9.67 = 61 \text{ rad s}^{-1}$$

$$E_k = \frac{1}{2} m \omega^2 (A^2 - y^2)$$

$$E_k = \frac{1}{2} \times 3.67 \times 61^2 \times (0.013^2 - 0)$$

$$\underline{E_k = 1.2 \text{ J}}$$

Worked example 4 – 2019 Q11(b)

A coin on the lid of the same tin loses contact when the tin's downward acceleration reaches 9.8 m s^{-2} . Find that displacement.

$$a = 9.8 \text{ m s}^{-2}$$

$$\omega = 61 \text{ rad s}^{-1}$$

$$y = ?$$

$$a = -\omega^2 y$$

$$y = a/\omega^2$$

$$y = 9.8/61^2$$

$$\underline{y = 2.6 \times 10^{-3} \text{ m}}$$



Damping

In a real system, friction transfers energy out of the system.

The total energy of an oscillator is $E_{\text{tot}} = \frac{1}{2}m\omega^2A^2$.

So as the energy falls, the amplitude of the motion falls too.

How quickly that happens is the damping of the system.



Underdamped, critically damped, overdamped

Underdamped: the system oscillates, and the amplitude dies away.

Critically damped: it does not oscillate at all, and it reaches rest in the shortest possible time.

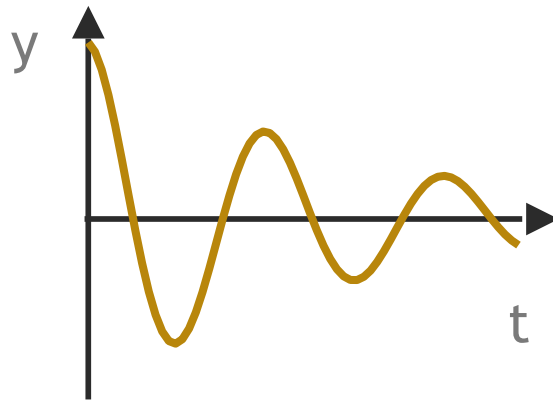
Overdamped: it does not oscillate either, but takes longer to come to rest than a critically damped system does.

Three kinds of damping

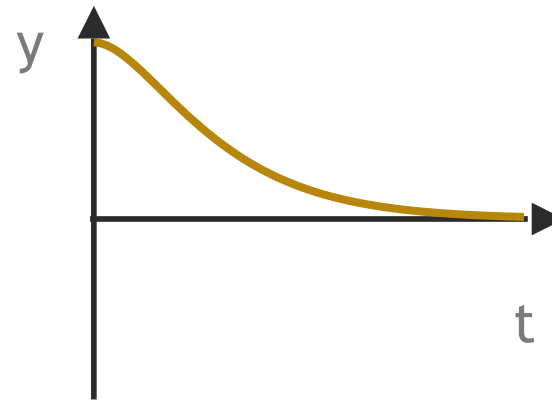


Critical damping reaches rest quickest, without oscillating

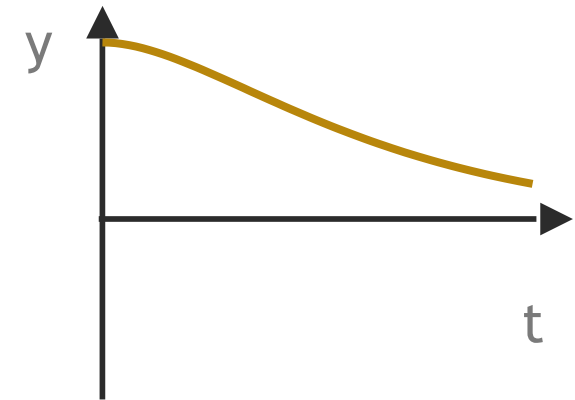
Underdamped



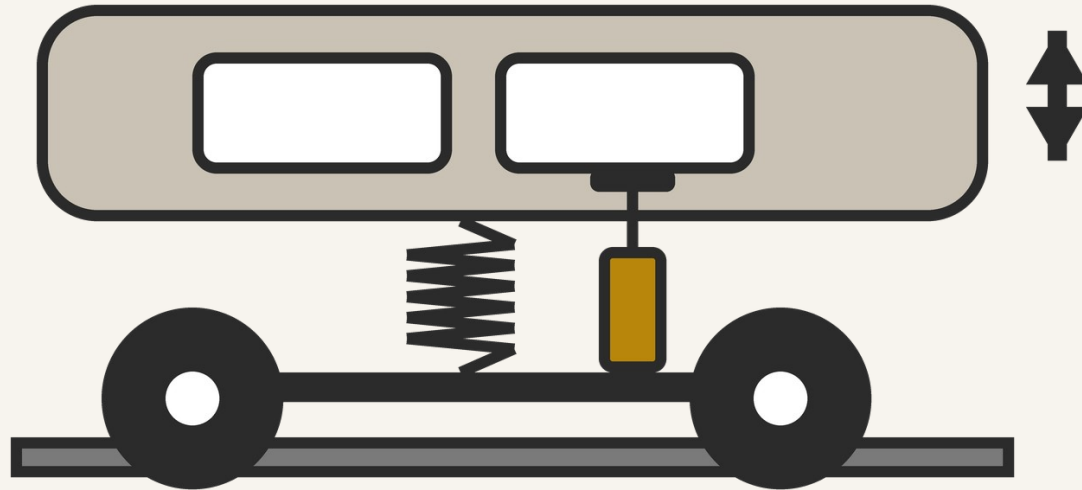
Critical



Overdamped



Why critical damping is the one you want



Worn shock absorbers underdamp, and the car keeps bouncing.

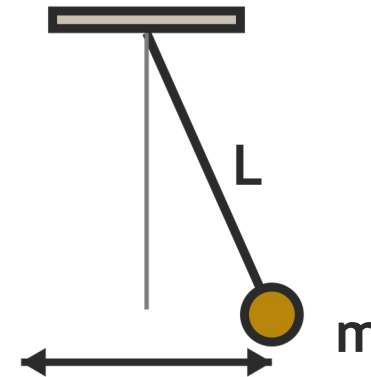
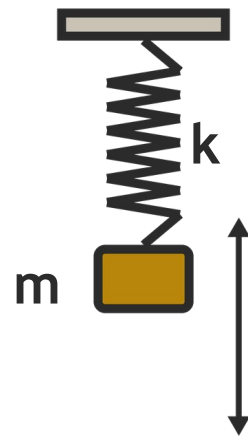
Two special cases



Not on the relationships sheet – the exam gives them to you

$$T = 2\pi\sqrt{m/k}$$

$$T = 2\pi\sqrt{L/g}$$



Each equation belongs to the system drawn beneath it.

SHM in the exam

A full question in every diet since 2016 – usually Q8 to Q11

Diet	Question	What it asked
2016	Q10	Define SHM; the cosine solution
2017	Q8	Choosing the right model
2020	Q9	Define SHM; the sine solution
2023	Q10(a), (b), (f)	Calculation
2024	Q9	Calculation
2025	Q9(b), (c)	Calculation

2018, 2019 and 2022 are worked in class above.

Reading and homework

Scholar: Quanta and waves, Topic 4 — Simple harmonic motion

Read the topic, then sit the Scholar end-of-topic test.

Scholar — Quanta and waves, Topic 4

Scholar — end-of-topic test (homework)

BrightRed Study Guide pp. 70-75

Past papers — 2016, 2017, 2020, 2023-25



Scholar

Check yourself

Can you do each of these without your notes?

- 1 State the definition of SHM, word for word.
- 2 Derive $a = -\omega^2 y$ from $y = A \sin \omega t$, showing every step.
- 3 Derive $v = \pm \omega \sqrt{A^2 - y^2}$, showing every step.
- 4 Derive $E_k = \frac{1}{2} m \omega^2 (A^2 - y^2)$, showing every step.
- 5 Describe underdamping, critical damping and overdamping.

Experiments you could try

Standard kit, or your own phone

- Mass on a spring with a motion sensor: plot T^2 against mass.
- Simple pendulum: plot T^2 against length, and find g .
- Trolley between two springs: take the y - t graph off a sensor.
- Loaded test tube bobbing in water: time 20 oscillations.
- Damping: add a card to the mass, then repeat it in water.
- DIY: a phone as accelerometer, or film the bob at 240 fps.

Sources & credits

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